# Sarthak Verma | EE23B104 | | | My Portfolio

**Indian Institute of Technology, Madras**


## Education

| Program | Institution | CGPA/% | Year of Completion |
|---|---|---|---|
| B.Tech in Electrical Engineering | Indian Institute of Technology Madras | 8.57/10 | 2027 |
| Std. XII (CBSE) | Holy Mission Secondary School, Patna | 96.8% | 2023 |
| Std. X (CBSE) | Loyola High School, Patna | 98.2% | 2021 |

## Scholastic Achievements

- Secured a position in **top 5** in the IMC Trading simulation, an algorithmic trading competition among 150+ teams.
- Secured a position in **top 30** in Optiver tradeathon with 300+ participants at IIT Madras.
- Achieved a respectable All India Rank in JEE Advanced & JEE Mains 2023 among more than 1.2 million candidates.
- Achieved a repectable rank in KVPY, 2022 conducted by the **IISC, Bangalore** among 50k+ aspirants.
- Recipient of the prestigious **NTSE** Fellowship by the **Govt of India** in the year 2021.
- Ranked in **National Top 1%** in NSEC 2022 and **State Top 1%** in NSEJS 2019, NSEP 2022 and NSEC 2022.
- **State Rank 1**, **Pythagoras Star** - State-level maths contest by **Connectech - UEM** and govt. collaboration.
- Achieved a **National Rank 3** in Montfort International Mathematics Olympiad, 2019.
- Secured an **International Rank of 28** and **Zonal Rank 3** in SOF International Mathematics Olympiad, 2020.
- Secured **Zonal Rank 1** and **International Rank 71** in Silverzone International Informatics Olympiad, 2019.

## Course Work & Skills

| | | |
|---|---|---|
| Stochastic Modelling and theory of Queues | Machine Learning Foundations | Deep Learning |
| Probability Foundations for Electrical Engineers | Linear Algebra | Multivariable Calculus |
| Advanced Game theory | Statistics for Electrical Engineers | Numerical Methods with C |

## Professional Experience

- **Quantitative Research Intern | Goldman Sachs** *May '26 - Jul '26*
  - Engineered calibration pipelines using **Stochastic Volatility Local Volatility (SVLV)** models for **KRW CMS - CMT** exotics, optimizing volatility surface fitting and front-office pricing accuracy.
  - Architected a market-benchmarking engine to validate exotic rate pricing against **Totem consensus data**, directly enhancing desk quoting precision and mark-to-market accuracy.
  - Developed a real-time risk & diagnostics engine to automate PnL explain, Greeks/risk attribution, and scenario decomposition, driving decision support for exotic rate traders.
- **CYBATHLON Competition | Zurich, Switzerland** *October 2024*
  - Represented IIT Madras and the **R2D2**** **Lab** at an international competition focused on assistive technologies.
  - Achieved a **global rank of 8**, marking the first Indian student-led team to compete in this domain.
  - Designed real-time control systems and embedded solutions for assistive devices focused on above-knee amputees.

## Research & Publications

- **Metaoptics - GUI-Driven Metaoptical Design Tool in Python** *Jan '25 - Present*
  - **Paper Submitted:** Co-authored research paper submitted to **Journal of Optics** (**Springer Nature**, official journal of the Optical Society of India) under the guidance of **Prof. Shanti Bhattacharya**.
  - Developed Python software contributing to fast and efficient generation of GDS files for Metamaterial fabrication.
  - Supported hexagonal lattices and **dual-wavelength phase profiling** via variable-radius nanopillars.

## Projects

- **Transformer-Based Stock Return Forecasting** *Jan '26 - Mar '26*
  - Developed a **Transformer forecasting model** in PyTorch to capture long-range dependencies in time series.
  - Engineered **price, volume, momentum, and volatility features** with sliding-window sampling, positional encoding, multi-head self-attention, and multi-horizon return forecasting using walk-forward validation.
- **Trading Strategy Arena - Quantitative Backtesting & Signal Engine** *May '25 - Jul '25*
  - Built an **event-driven backtesting engine** in Python to benchmark multi-agent strategies via Tkinter.
  - Modeled dynamic market frictions (slippage, TCA, execution latency) to eliminate systematic lookahead bias.
  - Evaluated systematic StatArb and multi-factor ML signals, tracking Sharpe ratio, Alpha/Beta, and rolling VaR.
- **Simulation and Optimization in Python (Course Project)** *Jul '24 - Nov '24*
  - Leveraged Python to build a **SPICE circuit simulator** and optimized topologies via **Simulated Annealing**.
  - Profiled bottlenecks and implemented **Cython** C-extensions to accelerate performance-critical execution routines.
- **Numerical Methods with C (Course Project)** *Jul '23 - Nov '23*
  - Implemented root-finding, Lagrangian/spline interpolation, Gaussian Elimination, and LU Decomposition solvers.
  - Solved non-linear **Landau-Lifshitz-Gilbert ODEs** via RK, Heun, and Euler-Cauchy for magnetic switching.
  - Generated pseudo-random sequences, computed Hamming distances, and validated randomness via NIST suite.

## Leadership & Entrepreneurship

- **LeetCore - Founder** *May '26 - Present*
  - Engineered and scaled an online assessment platform to **2,500+ active users** with proctored contest engines.
  - Formed a partnership with **GlobalFoundries** to integrate industry-aligned technical evaluation modules.
  - Deployed full-stack infrastructure at **leetcore.co.in** featuring real-time analytics and mock exam suites.
- **Head EE UG, Placement and Internship Cell** *May '26 – present*
  - Leading a team of 14 for the upcoming placement and internship for the EE department at IITM.
  - Coordinated with 20+ Companies and 40+ Profiles as Primary/Secondary POC during the previous season.
  - Developed and implemented placement policies and guidelines for students and recruiters at IIT Madras.

** Rehabilitation Research and Device Development